2026

Project Proposal

ONLINE LEARNING PLATFORM

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TO | Mr. Ihtisham Ullah

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Database Management System (DBMS) Semester Project Proposal

Project Title:

Online Learning Platform (LMS with Analytics)

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Class:

BSCS 4-2

Semester:

Spring 2026

1. Introduction

With the rapid growth of digital education, Learning Management Systems (LMS) have become essential in modern academic environments. Many institutions lack a centralized system to efficiently manage courses, student enrollments, assignments, and performance tracking.

The **Online Learning Platform (LMS with Analytics)** is a database-driven system designed to manage academic operations such as user management, course handling, enrollments, and assessments.

Problem Description:
Manual or poorly structured systems lead to data redundancy, inconsistency, and difficulty in tracking student performance.

Importance:
This system improves academic management, enhances data organization, and provides analytical insights for better decision-making.

Use of DBMS Concepts:
This project applies core DBMS concepts including:

- Entity-Relationship Modeling
- Normalization (up to 3NF)
- Constraints (PK, FK, etc.)
- Advanced SQL Queries

2. Objectives

- Design a structured and efficient relational database
- Ensure data integrity using constraints
- Apply normalization up to Third Normal Form (3NF)
- Implement complex SQL queries (joins, subqueries, aggregation)
- Generate analytical reports (student performance, course trends)

3. Scope of the Project

Target Users:

- Students
- Instructors
- Administrators

System Boundaries:

- Focus only on database design and implementation
- No front-end or UI development included

Key Features:

- User management
- Course management
- Student enrollments
- Assignment handling and submissions
- Performance tracking and analytics

4. Entity Relationship Overview (ERD + Diagram)

Entity 1: Users

- user_id (PK)
- name
- email (UNIQUE)
- role (Student/Instructor/Admin)

Entity 2: Courses

- course_id (PK)
- title
- description
- instructor_id (FK)

Entity 3: Enrollments

- enrollment_id (PK)
- user_id (FK)
- course_id (FK)
- enrollment_date

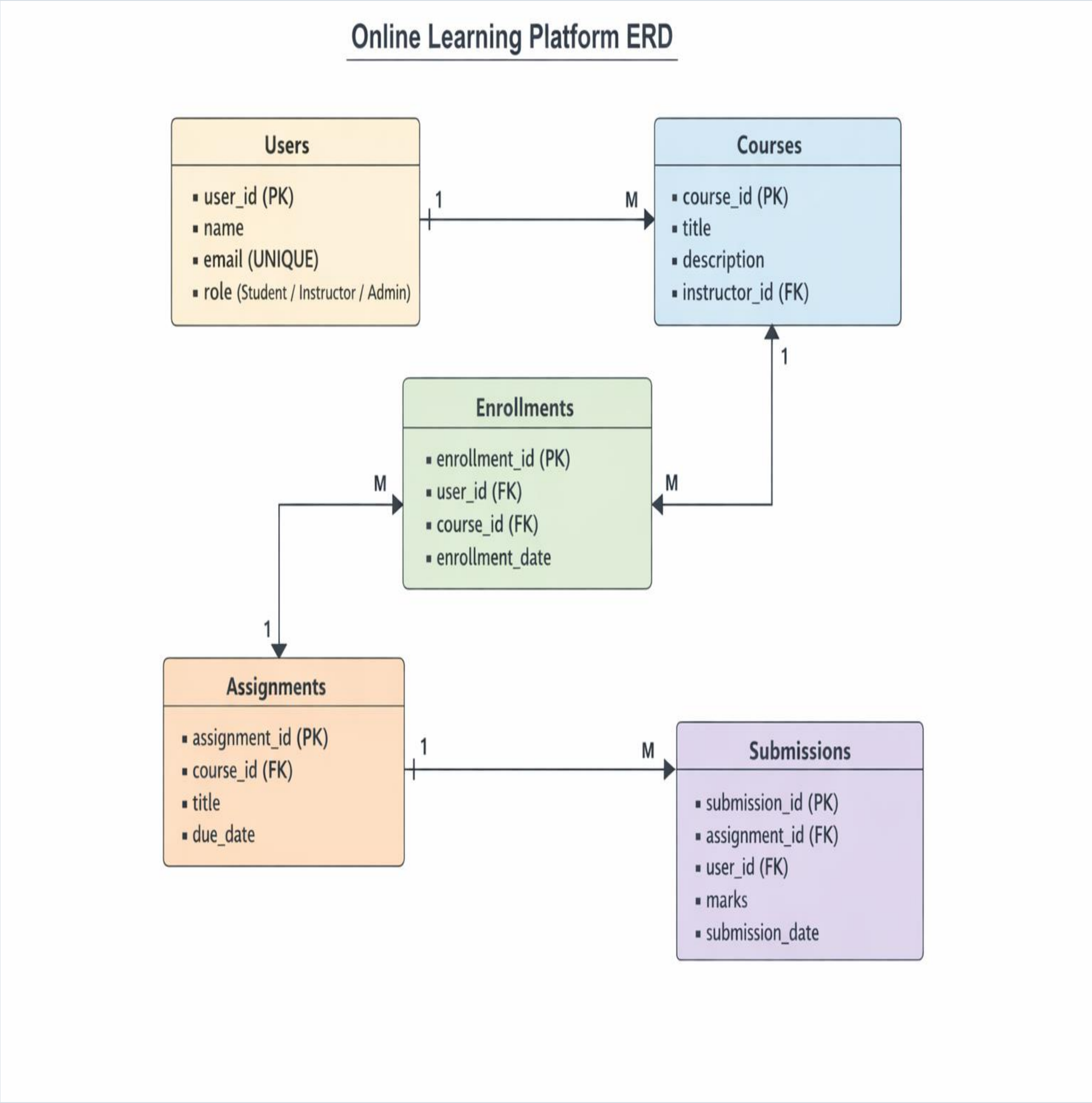
Entity 4: Assignments

- assignment_id (PK)
- course_id (FK)
- title
- due_date

Entity 5: Submissions

- submission_id (PK)
- assignment_id (FK)
- user_id (FK)
- marks
- submission_date

ERD-DIAGRAM



Online Learning Platform ERD
Users
user_id (PK)
name
email (UNIQUE)
role (Student / Instructor / Admin)
Courses
course_id (PK)
title
description
instructor_id (FK)
Enrollments
enrollment_id (PK)
user_id (FK)
course_id (FK)
enrollment_date
Assignments
assignment_id (PK)
course_id (FK)
title
due_date
Submissions
submission_id (PK)
assignment_id (FK)
user_id (FK)
marks
submission_date
1
M
1
M
M
1
1
M

5. Database Design (Schema)

Primary Keys:

- user_id, course_id, enrollment_id, assignment_id, submission_id

Foreign Keys:

- Courses.instructor_id → Users.user_id
- Enrollments.user_id → Users.user_id
- Enrollments.course_id → Courses.course_id
- Assignments.course_id → Courses.course_id
- Submissions.assignment_id → Assignments.assignment_id
- Submissions.user_id → Users.user_id

Relationships:

- One User → Many Courses (Instructor role)
- Many Users ↔ Many Courses (via Enrollments)
- One Course → Many Assignments
- One Assignment → Many Submissions

6. Normalization

1NF:

- All attributes are atomic (no repeating groups)

2NF:

- No partial dependency (all attributes depend on full PK)

3NF:

- No transitive dependency (non-key attributes depend only on PK)

Benefits:

- Reduces redundancy
- Improves data consistency
- Enhances database efficiency

7. Constraints

- **Primary Key** → Unique identification of records
- **Foreign Key** → Maintain referential integrity
- **NOT NULL** → Mandatory fields (e.g., name, course title)
- **UNIQUE** → Email uniqueness
- **CHECK** → Marks ≥ 0

8. Advanced SQL Queries

- **Joins:** Retrieve student-course details
- **Subqueries:** Find top-performing students
- **Aggregation:** Average marks, course enrollments
- **Filtering:** Students with marks above threshold
- **Reports:**
    - Student performance report
    - Course popularity report

9. Expected Outputs

- Student performance reports
- Course enrollment summaries
- Assignment submission records
- Analytical summaries (average marks, course trends)

10. Tools & Technology

- **Database:** Oracle Database
- **Language:** SQL / PL-SQL
- **Tools:** Oracle SQL Developer

11. Conclusion

The proposed LMS with Analytics is a structured and practical database project that demonstrates strong understanding of DBMS concepts. By limiting the system to five main entities, the design remains efficient while still supporting meaningful operations and analytics. This project will provide hands-on experience in database design, normalization, and advanced SQL querying.